

POINT OF AYRE LH, Isle of Man

WMO: 032080

Lat: 54.417N

Lon: 4.367W

Elev: 30

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 86-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.1	34.8	24.0	18.2	35.9	26.2	20.2	36.8	47.9	43.4	44.0	45.3	13.6	120	0.794

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	68.5	61.2	66.6	60.1	65.1	59.2	62.8	66.4	61.5	65.0	60.3	63.7	12.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	81.1	64.6	60.0	77.3	63.2	58.7	73.8	62.3	28.4	66.5	27.5	64.8	26.6	64.0	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.2	33.4	30.4	DB	28.9	74.4	2.9	2.3	26.8	76.1	25.1	77.4	23.5	78.7	21.4	80.4	(4)
(5)			WB	27.4	66.0	3.7	1.5	24.8	67.1	22.6	68.0	20.5	68.8	17.8	69.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	42.7	42.5	44.2	46.4	51.7	55.7	59.3	59.8	56.9	52.7	47.6	44.3	(6)
(7)		DBStd	7.24	4.25	4.36	3.94	3.44	3.64	3.49	2.72	3.05	3.38	4.04	3.72	4.14	(7)
(8)		HDD50	1054	229	211	183	116	23	1	0	0	1	20	89	181	(8)
(9)		HDD65	5346	692	629	645	558	411	279	179	165	242	382	522	642	(9)
(10)		CDD50	1188	1	2	4	8	77	173	287	302	209	103	17	5	(10)
(11)		CDD65	4	0	0	0	0	0	1	1	2	0	0	0	0	(11)
(12)		CDH74	4	0	0	0	0	0	1	1	2	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	15.8	19.2	19.4	17.5	14.2	13.3	12.9	12.7	12.8	14.3	17.3	18.1	18.7	(14)
(15)	Precipitation	PrecAvg	33.0	3.3	2.3	2.4	2.1	1.9	1.9	2.0	2.8	3.0	3.9	3.8	3.5	(15)
(16)		PrecMax	39.5	7.3	5.1	4.7	4.2	5.4	4.2	5.1	5.5	5.4	7.2	6.4	6.5	(16)
(17)		PrecMin	26.3	0.6	0.0	0.6	0.3	0.4	0.6	0.5	0.3	0.3	0.6	1.1	0.5	(17)
(18)		PrecStd	3.6	1.4	1.3	1.1	1.1	1.2	0.9	1.2	1.2	1.3	1.8	1.3	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	54.9	55.4	58.5	65.0	70.5	71.4	71.1	68.3	63.5	58.1	56.3	(19)
(20)			MCWB	51.1	50.9	51.0	53.2	57.7	61.4	62.3	63.5	61.4	57.0	55.3	54.1	(20)
(21)		2%	DB	52.1	52.8	53.2	56.0	61.9	66.3	68.3	68.2	65.4	61.2	56.0	53.6	(21)
(22)			MCWB	49.5	49.4	49.8	51.2	55.8	58.5	61.1	61.6	58.8	56.7	53.5	51.8	(22)
(23)		5%	DB	50.6	51.0	51.5	53.7	59.6	63.6	66.3	66.4	63.6	59.9	54.7	51.9	(23)
(24)			MCWB	48.4	48.5	48.8	49.6	54.1	57.4	59.8	60.7	57.6	56.0	52.4	50.1	(24)
(25)		10%	DB	49.1	49.2	50.0	52.0	57.7	61.6	64.5	64.9	62.0	58.4	53.3	50.4	(25)
(26)			MCWB	47.1	46.8	47.5	48.6	52.8	56.4	58.9	59.9	57.1	55.0	51.2	48.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.0	52.0	54.4	58.9	62.2	64.5	65.1	62.9	59.6	55.9	54.5	(27)
(28)			MCD B	53.1	53.1	54.5	58.2	64.3	68.7	69.2	69.1	65.6	61.2	57.3	55.9	(28)
(29)		2%	WB	50.2	49.7	50.5	52.2	56.6	59.7	62.1	63.3	60.9	58.3	54.1	52.1	(29)
(30)			MCD B	51.8	51.7	52.4	55.0	60.8	64.6	66.8	66.3	63.6	60.1	55.3	53.2	(30)
(31)		5%	WB	48.7	48.4	49.2	50.7	54.9	58.4	60.8	62.0	59.4	56.9	52.8	50.7	(31)
(32)			MCD B	50.3	50.2	51.1	53.2	58.5	62.6	65.0	64.8	62.0	58.8	54.1	51.8	(32)
(33)		10%	WB	47.4	46.7	47.8	49.2	53.6	57.1	59.6	60.9	58.2	55.7	51.5	48.9	(33)
(34)			MCD B	49.1	48.5	49.7	51.6	57.0	60.8	63.4	63.8	60.9	57.7	53.0	50.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	6.5	7.1	8.4	9.8	9.4	9.2	8.9	8.1	6.3	5.6	5.3	(35)
(36)			MCDBR	7.7	7.8	8.5	11.1	12.4	12.6	12.4	10.8	9.6	7.9	7.1	6.4	(36)
(37)			MCWBR	7.7	7.3	6.7	7.8	7.4	7.3	6.8	6.4	6.3	6.0	7.1	6.9	(37)
(38)		5% WB	MCD BR	7.4	7.9	8.1	10.3	11.7	11.6	11.2	9.9	8.7	6.8	7.0	6.5	(38)
(39)			MCWBR	7.6	8.0	7.0	7.9	7.8	7.6	7.0	6.7	7.1	6.3	7.4	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.344	0.369	0.387	0.391	0.385	0.387	0.373	0.365	0.350	0.327	0.319	(40)	
(41)		taud	2.280	2.363	2.336	2.320	2.367	2.412	2.410	2.463	2.482	2.452	2.358	2.259	(41)	
(42)		Ebn at Noon	189	234	254	266	271	274	271	268	255	230	193	167	(42)	
(43)		Edh at Noon	19	24	31	36	36	35	34	31	27	22	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	179	396	774	1247	1600	1621	1546	1277	888	485	229	133	(44)	
(45)		RadStd	19	40	61	94	130	154	132	94	59	51	24	12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page